

SALES ANALYSIS

Empower Businesses to make Data-Driven Decisions

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GET TO KNOW ABOUT DATA



Sales Analysis Dataset Overview

This dataset contains information about the company's sales transactions, focusing on factors that impact revenue and daily operations. It includes numerical and categorical data describing our products, physical stores, purchasing channels, and financial metrics. By analyzing these columns, we can identify key trends, evaluate our profit margins, and make smarter decisions to grow the business.

DEFINING THE COLUMNS AND DATA TYPES



Overall Column Overview

The dataset contains a comprehensive set of features related to the company's sales transactions, focusing on various factors that can influence revenue and business performance. It includes product hierarchies, geographic locations, store characteristics, and financial metrics such as costs, discounts, and profit margins.

By analyzing these columns, we aim to identify patterns that help in tracking sales trends, evaluating regional success, and making data-driven decisions to maximize profitability.

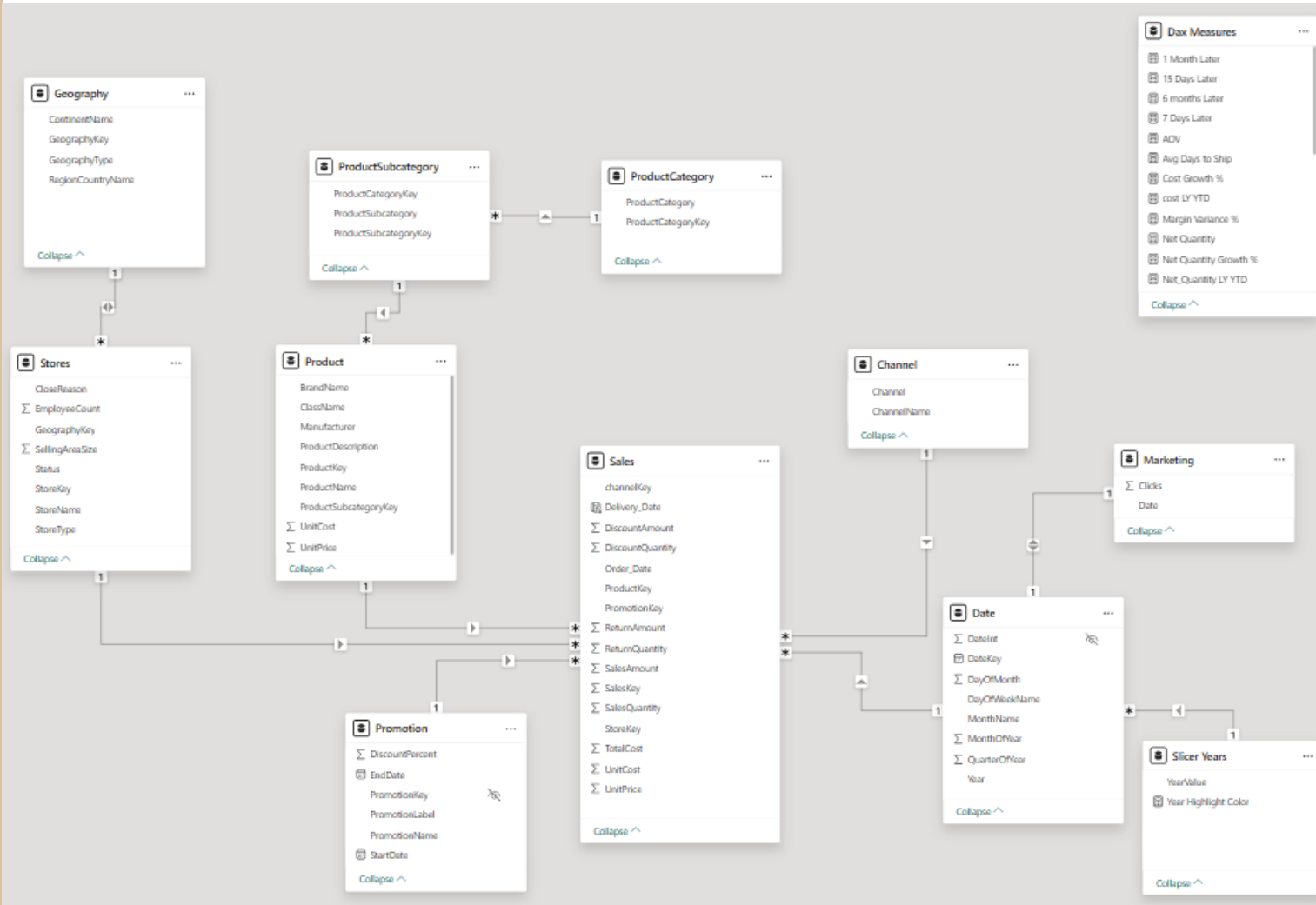
DEFINING THE COLUMNS AND DATA TYPES

- **BrandName:** Identifies the manufacturer associated with each product (e.g., Contoso).
- **ChannelName:** The purchasing platform used by the customer (Physical Store, Online, Catalog, or Reseller).
- **ClassName:** Categorizes the product's market segment (e.g., Economy, Regular, Premium).
- **Clicks:** Tracks daily customer engagements and interactions originating from digital marketing campaigns.
- **ContinentName:** Broad geographic classification for global market analysis.
- **Delivery_Date:** The exact date when the purchased items were delivered to the customer.
- **DiscountAmount:** The monetary value deducted from the original price due to applied promotions.
- **DiscountPercent:** The specific discount rate applied to the sale through marketing campaigns.
- **EmployeeCount:** The number of staff members assigned to a specific physical store.
- **Order_Date:** The specific date when the transaction occurred, essential for tracking time-based trends.
- **ProductCategory:** Broad classification of products to analyze high-level sales trends.

DEFINING THE COLUMNS AND DATA TYPES

- **ProductName:** Represents the specific name and model of the product being sold.
- **ProductSubcategory:** Detailed classification under the main category (e.g., Laptops under Computers).
- **PromotionName:** The specific marketing campaign or holiday offer applied to the transaction.
- **RegionCountryName:** The specific country where the sale took place, essential for tracking regional success.
- **ReturnQuantity:** The number of units returned by customers, impacting the net revenue.
- **SalesAmount:** The final calculated revenue for the entire transaction after all discounts.
- **SalesQuantity:** The number of units purchased in a single transaction.
- **SellingAreaSize:** The physical size of the store, used to evaluate retail footprint efficiency.
- **StoreName:** The physical location where the retail transaction took place.
- **TotalCost:** The total incurred cost for the entire transaction.
- **UnitCost:** The base cost required to manufacture or acquire a single unit of the product.
- **UnitPrice:** The official selling price of the product before any applied discounts.

BUILDING A SCHEMA



BUILDING A SCHEMA

- This data model is designed to analyze company sales and revenue, focusing on various factors that influence profitability. It uses a snowflake schema, with a central Fact Table and multiple Dimension Tables:
- **Fact_Sales:** The central table containing key metrics related to transactions, such as Sales Quantity, Total Cost, Return Amount, and Sales Amount.
- **Dimension Tables:**
 - **Dim_Product:** Contains product details, categories, and brand names.
 - **Dim_Geography:** Categorizes locations by city, country, and continent.
 - **Dim_Stores:** Describes physical store attributes and selling area size.
 - **Dim_Promotion:** Represents active marketing campaigns and discounts.
 - **Dim_Channel:** Lists purchasing mediums (Store, Online, Catalog).
 - **Dim_Date:** Time intelligence table for tracking daily operations.
- **Relationships:** These tables are connected via foreign keys (e.g., "ProductKey," "StoreKey," and "DateKey") to enable detailed analysis, such as sales by region, product performance, and marketing success.
- The purpose of this model is to help analysts identify patterns and factors that contribute to revenue growth by examining the relationships between these dimensions and key sales metrics.

BUILDING A SCHEMA

The data model I built incorporates several custom attributes and optimization techniques to enhance analysis and improve performance:

- **Custom Slicer Tables (DAX):**

For better dashboard interactivity, I created a custom table called "Slicer Years" using the `SELECTCOLUMNS` DAX function. This provides a dedicated, optimized filter for time-based analysis without cluttering the main dimension tables.

- **Hierarchical Categorization:**

The dataset is structured with natural hierarchies. Products are grouped from broad Categories to Subcategories and specific Product Names. Similarly, locations are mapped from Continents down to Countries and Cities, allowing for granular drill-down analysis.

- **Indexing for Faster Joins:**

To optimize performance, I avoided using text strings for the relationships between the fact and dimension tables. Instead, I utilized integer indexes (Keys) such as `SalesKey`, `ProductKey`, and `GeographyKey`. This method enhances the speed of joins and ensures more efficient querying for millions of rows.

By using these custom tables and indexing strategies, the model ensures more efficient data processing and better categorization, which leads to faster and more insightful analysis of sales performance.

MEASURES USING VARIABLES

The measures are designed to calculate specific performance and financial metrics based on the data in the fact and dimension tables. Here's a breakdown of the core calculations:

- **Core Sales & Cost Aggregations:**

These foundational measures calculate the absolute totals for revenue, cost, and profitability across the dataset.

```
Total Sales = SUM(Sales[SalesAmount])
Total Cost = SUM(Sales[TotalCost])
Total Profit = [Total Sales] - [Total Cost]
Total Quantity = SUM(Sales[SalesQuantity])
total return amount = SUM(Sales[ReturnAmount])
Total Return Quantity = SUM(Sales[ReturnQuantity])
Total Revnue = [Total Sales] - [total return amount]
Net Quantity = [Total Quantity] - [Total Return Quantity]
```

- **Key Performance Ratios:**

Calculates essential efficiency metrics such as Average Order Value (AOV) and profit margins.

```
Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)
Return Rate % = DIVIDE([Total Return Quantity], [Total Quantity], 0)
AOV = DIVIDE([Total Sales], COUNTROWS('Sales'), 0)
```

MEASURES USING VARIABLES

- **Global Sales & Revenue Contribution (%):**

Evaluates the percentage of sales generated by a specific filter context (e.g., country) against the global total using variable scoping.

```
sales % =  
VAR CountrySales = [Total Sales]  
VAR TotalGlobalSales =  
    CALCULATE(  
        [Total Sales],  
        ALL('Sales')  
    )  
RETURN  
    DIVIDE(CountrySales, TotalGlobalSales, 0)
```

```
Revenue % =  
VAR CountryRevenue = [Total Revenue]  
VAR TotalGlobalRevenue =  
    CALCULATE(  
        [Total Revenue],  
        ALL('Sales')  
    )  
RETURN  
    DIVIDE(CountryRevenue, TotalGlobalRevenue, 0)
```

- **Average Days to Ship:**

Measures logistics efficiency by calculating the average time taken between order placement and delivery.

```
Avg Days to Ship =  
AVERAGEX(  
    'Sales',  
    DATEDIFF('Sales'[Order_Date], 'Sales'[Delivery_Date], DAY)  
)
```

MEASURES USING VARIABLES

- Time Intelligence (LY Comparisons):

These measures use advanced variables to dynamically fetch the performance metrics from the exact same period in the previous year, ensuring reliable year-over-year benchmarking.

```
Sales LY =  
VAR LastAvailableYear = YEAR(MAX('Sales'[Order_Date]))  
VAR SelectedLY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[DateKey]))  
VAR DefaultLY = CALCULATE(  
    [Total Sales],  
    FILTER(ALL('Date'), 'Date'[Year] = LastAvailableYear - 1)  
)  
RETURN  
IF(  
    ISFILTERED('Date'[Year]),  
    COALESCE(SelectedLY, 0),  
    COALESCE(DefaultLY, 0)  
)
```

```
cost LY YTD =  
VAR LastAvailableYear = YEAR(MAX('Sales'[Order_Date]))  
VAR SelectedLY = CALCULATE([Total Cost], SAMEPERIODLASTYEAR('Date'[DateKey]))  
VAR DefaultLY = CALCULATE(  
    [Total Cost],  
    FILTER(ALL('Date'), 'Date'[Year] = LastAvailableYear - 1)  
)  
RETURN  
IF(ISFILTERED('Date'[Year]), COALESCE(SelectedLY, 0), COALESCE(DefaultLY, 0))
```

MEASURES USING VARIABLES

- **Growth & Variance Analysis (%):**

Calculates the percentage increase or decrease in various metrics compared to historical data to highlight business trends.

```
Sales Growth % =  
VAR CurrentSales = [Total Sales]  
VAR LastYearSales =  
    CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[DateKey]))  
RETURN  
    IF(  
        ISBLANK(LastYearSales), 0,  
        DIVIDE(CurrentSales - LastYearSales, LastYearSales, 0)  
    )
```

```
Cost Growth % =  
VAR CurrentCost = [Total Cost]  
VAR LastYearCost = CALCULATE([Total Cost], SAMEPERIODLASTYEAR('Date'[DateKey]))  
RETURN  
    DIVIDE(CurrentCost - LastYearCost, LastYearCost, 0)
```

```
Margin Variance % =  
VAR CurrentMargin = [Profit Margin %]  
VAR LastYearMargin = [Profit Margin LY YTD]  
RETURN  
    IF(  
        ISBLANK(CurrentMargin) || ISBLANK(LastYearMargin),  
        0,  
        CurrentMargin - LastYearMargin  
    )
```

MEASURES USING VARIABLES

- **Delivery Tracking (Post-Order Context):**

These measures calculate the volume of sales delivered within specific timeframes after the selected date, helping to track logistical bottlenecks.

```
15 Days Later =  
VAR SelectedDate = SELECTEDVALUE('Date'[DateKey])  
RETURN  
CALCULATE(COUNTROWS('Sales'),  
'Sales'[Delivery_Date] > SelectedDate && 'Sales'[Delivery_Date] <= SelectedDate + 15)  
  
1 Month Later =  
VAR SelectedDate = SELECTEDVALUE('Date'[DateKey])  
RETURN  
CALCULATE(COUNTROWS('Sales'),  
'Sales'[Delivery_Date] > SelectedDate && 'Sales'[Delivery_Date] <= SelectedDate + 30)
```

- **Chart Specific Overrides:**

These measures intentionally remove the context filters from the Slicer Years table to allow full historical trends to be displayed in specific visuals.

```
Sales for Chart =  
CALCULATE([Total Sales], REMOVEFILTERS('Slicer Years'))  
  
Profit for Chart =  
CALCULATE([Total Profit], REMOVEFILTERS('Slicer Years'))
```

BUILDING DYNAMIC INTERACTIVE DASHBOARD



The Power BI dashboard is designed to analyze company sales performance by leveraging custom DAX measures and advanced interactive features. To ensure a highly professional and user-friendly interface, the dashboard's background and layout were creatively custom-designed using Figma before being imported into Power BI.

Key metrics such as Total Revenue, Profit Margin, and Sales Growth are displayed across various intuitive visualizations. Interactive elements, including the custom "Slicer Years" and geographical filters, allow stakeholders to seamlessly drill down into specific brands, regions, and purchasing channels. This dynamic and aesthetically pleasing design enables executives and analysts to uncover hidden trends, evaluate operational efficiency, and ultimately make smarter, data-driven business decisions.